

Investigation of the Continuum Limit in the Presence of Small Breaking of Gauge Invariance in Lattice Gauge Theories

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May 18, 2026

This thesis investigates whether and how the continuum limit of a lattice gauge theory is affected by small explicit violations of gauge invariance. The study [...] numerical measurements of Wilson loops, Creutz ratios, the static potential, the Sommer scale, and related scaling observables. The central goal is to determine whether gauge-breaking effects vanish, persist, or modify the extrapolation toward the continuum limit as the lattice spacing is reduced.

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Notation and Conventions

a Lattice spacing.

β Bare lattice coupling parameter.

ε Parameter controlling the explicit breaking of gauge invariance.

$W(R, T)$ Wilson loop with spatial extent R and temporal extent T .

$\chi(R, T)$ Creutz ratio used to estimate the force and string tension.

r_0 Sommer scale defined through $r^2 F(r) = 1.65$.

σ String tension.

1. Introduction

1.1. Motivation

Explain why gauge invariance is normally central in lattice gauge theory and why controlled violations are relevant. Possible motivations include numerical robustness, effective descriptions, regulator artifacts, quantum simulation errors, or imperfect implementation of gauge constraints.

1.2. Research Question

State the main question precisely:

Does a small explicit breaking of gauge invariance disappear in the continuum limit, or does it alter the continuum extrapolation of physical observables?

1.3. Strategy and Main Observables

Introduce the numerical strategy and the observables used to probe the continuum limit: Wilson loops, Creutz ratios, the static potential $V(R)$, r_0 , and σ .

2. Theoretical Background

The aim of this chapter is to collect the theoretical ingredients needed for the analysis of the continuum limit in the presence of small explicit violations of gauge invariance. The discussion starts from continuum Yang–Mills theory and then introduces the lattice formulation, Wilson loops, the static potential, Creutz ratios, and the Sommer scale. These are the central concepts used later to define and compare physical observables at different lattice spacings.

2.1. Gauge Theories in the Continuum

2.1.1. Gauge Fields and Local Gauge Symmetry

A non-Abelian gauge theory is built from fields that transform locally under a Lie group. In this thesis the relevant case is the group $SU(2)$. A continuum gauge field can be written as

$$A_\mu(x) = A_\mu^a(x)T^a, \quad (2.1)$$

where T^a are the generators of the Lie algebra of $SU(2)$. A common choice is $T^a = \sigma^a/2$, with σ^a the Pauli matrices. The gauge field defines the covariant derivative

$$D_\mu = \partial_\mu + iA_\mu, \quad (2.2)$$

and the associated field-strength tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]. \quad (2.3)$$

The commutator term distinguishes a non-Abelian gauge theory from an Abelian one. It implies that the gauge field itself carries charge under the gauge group and therefore self-interacts.

A local gauge transformation is specified by a group element

$$\Omega(x) \in SU(2) \quad (2.4)$$

at every point in space-time. Under such a transformation, matter fields in the fundamental representation transform as

$$\psi(x) \mapsto \Omega(x)\psi(x), \quad (2.5)$$

while the gauge field transforms inhomogeneously,

$$A_\mu(x) \mapsto \Omega(x)A_\mu(x)\Omega^\dagger(x) - i(\partial_\mu\Omega(x))\Omega^\dagger(x). \quad (2.6)$$

The inhomogeneous derivative term is necessary because the transformation is local. The field strength transforms covariantly,

$$F_{\mu\nu}(x) \mapsto \Omega(x) F_{\mu\nu}(x) \Omega^\dagger(x), \quad (2.7)$$

so traces of products of field strengths at the same point are gauge invariant.

In Euclidean space-time, the pure Yang–Mills action can be written as

$$S_{\text{YM}} = \frac{1}{2g^2} \int d^4x \operatorname{tr} F_{\mu\nu}(x) F_{\mu\nu}(x), \quad (2.8)$$

with summation over repeated Lorentz indices. This action is invariant under local gauge transformations. The coupling g controls the strength of the interaction. For an asymptotically free gauge theory, the effective coupling becomes small at short distances, while the long-distance theory is strongly coupled and requires non-perturbative methods.

2.1.2. Gauge-Invariant Observables

Physical observables in a gauge theory must be gauge invariant. Gauge-dependent quantities can be useful after a gauge fixing has been imposed, but they do not directly correspond to physical observables unless the gauge dependence is properly controlled. For pure gauge theory, important gauge-invariant observables include traces of closed parallel transporters, local action-density operators, correlation functions of gauge-invariant operators, and quantities derived from Wilson loops.

A central continuum object is the path-ordered parallel transporter along a curve C ,

$$U(C) = \mathcal{P} \exp \left(i \int_C dx^\mu A_\mu(x) \right). \quad (2.9)$$

For a closed curve, the trace

$$W(C) = \frac{1}{N_c} \operatorname{Re} \operatorname{tr} U(C) \quad (2.10)$$

is gauge invariant. On the lattice this object becomes the Wilson loop and is one of the most direct probes of confinement and of the static quark–antiquark potential.

2.1.3. Physical Consequences of Gauge Invariance

Gauge invariance is not only a formal redundancy of the description. It constrains the allowed terms in the action, the form of correlation functions, and the possible continuum limits. In the lattice formulation, preserving gauge invariance exactly at nonzero lattice spacing is usually advantageous because it prevents gauge-noninvariant counterterms from being generated by radiative corrections. Wilson’s lattice gauge action realizes this idea by putting compact group-valued variables on links and by constructing the action from traces of closed loops [1, 2, 3].

The purpose of this thesis is to study what happens when this exact symmetry is slightly violated by an explicit deformation. From a renormalization-group point of view, the key question is whether the deformation behaves like an irrelevant lattice artifact, so that its effects vanish as $a \rightarrow 0$, or whether it defines a relevant perturbation that changes the continuum limit. This question is not purely formal: small violations of gauge constraints can appear as regulator artifacts, algorithmic imperfections, or implementation errors in effective or quantum-simulation formulations of lattice gauge theories [4, 5].

2.2. Lattice Gauge Theory

2.2.1. Discretization of Space-Time

In lattice gauge theory, Euclidean space-time is replaced by a hypercubic lattice

$$\Lambda_a = \{x = an \mid n \in \mathbb{Z}^4\}, \quad (2.11)$$

where a is the lattice spacing. The lattice spacing acts as an ultraviolet cutoff. Momenta larger than order $1/a$ are not resolved. The continuum limit is obtained by sending

$$a \rightarrow 0 \quad (2.12)$$

while keeping physical quantities fixed.

In a numerical simulation the lattice has finite extent

$$L_\mu = N_\mu a, \quad (2.13)$$

where N_μ is the number of lattice points in direction μ . Therefore two limits are conceptually distinct: the continuum limit $a \rightarrow 0$ and the infinite-volume limit $L_\mu \rightarrow \infty$. In practice, one tries to choose volumes large enough that finite-volume effects are smaller than the statistical and systematic errors relevant for the analysis. [in simulations in this thesis I use $L_{\mathbf{1}} = \text{const}$ for the space directions and $N_{\mathbf{1}} \text{ const}$ for time direction]

2.2.2. Link Variables and Plaquettes

The fundamental degrees of freedom of Wilson lattice gauge theory are link variables

$$U_\mu(x) \in \text{SU}(2), \quad (2.14)$$

which live on the oriented link from x to $x + a\hat{\mu}$. They are the lattice analogues of parallel transporters. For a smooth continuum field one may formally write

$$U_\mu(x) \approx \exp \left[iaA_\mu \left(x + \frac{a}{2}\hat{\mu} \right) \right]. \quad (2.15)$$

A local lattice gauge transformation assigns a group element $\Omega(x)$ to each site and acts on the link variables as

$$U_\mu(x) \mapsto \Omega(x)U_\mu(x)\Omega^\dagger(x + a\hat{\mu}). \quad (2.16)$$

The elementary closed loop on the lattice is the plaquette,

$$U_{\mu\nu}(x) = U_\mu(x)U_\nu(x + a\hat{\mu})U_\mu^\dagger(x + a\hat{\nu})U_\nu^\dagger(x). \quad (2.17)$$

Under a gauge transformation, the plaquette transforms covariantly at its base point,

$$U_{\mu\nu}(x) \mapsto \Omega(x)U_{\mu\nu}(x)\Omega^\dagger(x). \quad (2.18)$$

Therefore $\text{Re tr } U_{\mu\nu}(x)$ is gauge invariant.

For smooth fields, the plaquette has the continuum expansion

$$U_{\mu\nu}(x) = \exp \left[ia^2 F_{\mu\nu}(x) + \mathcal{O}(a^3) \right]. \quad (2.19)$$

Thus the plaquette is the lattice object from which the continuum field strength is recovered. [but not really used in this thesis, so maybe remove this point]

2.2.3. Wilson Gauge Action

The standard Wilson gauge action for $\text{SU}(N_c)$ is

$$S_W[U] = \beta \sum_x \sum_{\mu < \nu} \left(1 - \frac{1}{N_c} \text{Re tr } U_{\mu\nu}(x) \right), \quad (2.20)$$

with

$$\beta = \frac{2N_c}{g_0^2}. \quad (2.21)$$

For $\text{SU}(2)$ this becomes

$$\beta = \frac{4}{g_0^2}, \quad (2.22)$$

and

$$S_W[U] = \beta \sum_x \sum_{\mu < \nu} \left(1 - \frac{1}{2} \text{Re tr } U_{\mu\nu}(x) \right). \quad (2.23)$$

The action is built only from traces of closed loops and is therefore exactly gauge invariant at nonzero lattice spacing.

The continuum limit of the Wilson action is obtained by inserting the plaquette expansion. Up to an irrelevant additive constant and lattice artifacts,

$$S_W[U] \longrightarrow \frac{1}{2g_0^2} \int d^4x \text{tr } F_{\mu\nu}(x)F_{\mu\nu}(x). \quad (2.24)$$

This reproduces the Euclidean Yang–Mills action. The corrections at finite lattice spacing are cutoff effects. For the pure Wilson gauge action, the leading gauge-invariant lattice artifacts in many on-shell quantities are of order a^2 , although the actual scaling observed in numerical data can be affected by statistics, finite-volume effects, operator definitions, and the accessible range of lattice spacings.

2.2.4. Path Integral and Expectation Values

The lattice path integral is finite-dimensional at finite lattice spacing and finite volume. The partition function is

$$Z = \int \prod_{x,\mu} dU_\mu(x) e^{-S[U]}, \quad (2.25)$$

where $dU_\mu(x)$ is the Haar measure on $SU(2)$. The expectation value of an observable $\mathcal{O}[U]$ is

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \prod_{x,\mu} dU_\mu(x) \mathcal{O}[U] e^{-S[U]}. \quad (2.26)$$

Gauge-invariant expectation values can be computed without gauge fixing. This is one of the practical advantages of the compact Wilson formulation.

In Monte Carlo simulations the path integral is estimated by generating an ensemble of gauge-field configurations distributed according to $e^{-S[U]}$. For an ensemble of N_{cfg} configurations U_i , the estimator is

$$\langle \mathcal{O} \rangle \approx \frac{1}{N_{\text{cfg}}} \sum_{i=1}^{N_{\text{cfg}}} \mathcal{O}[U_i]. \quad (2.27)$$

Because successive configurations in a Markov chain are correlated, autocorrelations must be considered when estimating uncertainties.

2.2.5. Continuum Limit and Renormalization

The continuum limit is not obtained by simply decreasing a at fixed bare coupling. Instead, the bare parameters have to be tuned so that physical observables remain fixed in physical units. For pure $SU(2)$ gauge theory with the Wilson action, decreasing the lattice spacing corresponds to increasing β , or equivalently decreasing the bare coupling g_0 .

A useful way to formulate the continuum limit is in terms of a physical length scale, such as the Sommer scale r_0 . The dimensionless ratio

$$\frac{r_0}{a} \quad (2.28)$$

is measured on the lattice. Increasing r_0/a means that the same physical length is resolved by more lattice points, i.e. the lattice spacing is smaller in physical units. Therefore continuum extrapolations can be expressed as fits in

$$\left(\frac{a}{r_0} \right)^2 \quad (2.29)$$

or another measured dimensionless lattice-spacing proxy.

Continuum extrapolations compare observables at fixed physical conditions, not merely at fixed lattice indices. For observables derived from Wilson loops this means that a loop size R on a coarse ensemble and the same integer R on a fine ensemble do not represent

the same physical distance. Therefor instead compare at fixed ratios such as r/r_0 or fixed dimensionless distances, a reference scale is used. [This point is emphasized in recent work on controlled continuum extrapolations of Wilson-loop observables, where Creutz ratios are used as finite continuum quantities and fitted at fixed physical distance [6].

The renormalization-group interpretation is central for the present thesis. Different lattice actions can lead to the same continuum theory if they differ only by irrelevant operators. In that case they lie in the same universality class. By contrast, a relevant deformation changes the long-distance theory unless its coefficient is tuned to zero. [A gauge-breaking term must therefore be classified by its effect on long-distance, gauge-invariant observables as $a \rightarrow 0$. (not written well yet)] The numerical analysis in later chapters tests this question directly.

2.3. Static Potential and Confinement

2.3.1. Wilson Loops

A Wilson loop is the trace of the ordered product of link variables around a closed contour. For a rectangular loop in the $\mu\nu$ -plane with side lengths L_μ and L_ν in lattice units, the code measures the normalized real trace

$$W_{\mu\nu}(L_\mu, L_\nu) = \frac{1}{N_c V} \sum_x \text{Re tr} \left[U_\mu(x; L_\mu) U_\nu(x + L_\mu \hat{\mu}; L_\nu) U_\mu^\dagger(x + L_\nu \hat{\nu}; L_\mu) U_\nu^\dagger(x; L_\nu) \right], \quad (2.30)$$

where V is the lattice volume and

$$U_\mu(x; L_\mu) = U_\mu(x) U_\mu(x + \hat{\mu}) \cdots U_\mu(x + (L_\mu - 1)\hat{\mu}) \quad (2.31)$$

denotes a straight Wilson line of length L_μ in direction μ . The factor $1/N_c$ normalizes the trace, while the factor $1/V$ averages over all possible base points of the loop.

For temporal Wilson loops, the last lattice direction is used as the Euclidean time direction,

$$t = N_d - 1. \quad (2.32)$$

The code computes rectangular loops in each spatial-time plane and averages over the spatial directions:

$$W(R, T) = \frac{1}{N_d - 1} \sum_{\mu=0}^{N_d-2} W_{\mu,t}(R, T). \quad (2.33)$$

For the SU(2) simulations considered in this thesis, $N_c = 2$, so the stored temporal Wilson loop is

$$W(R, T) = \frac{1}{2V(N_d - 1)} \sum_{\mu=0}^{N_d-2} \sum_x \text{Re tr} W_{\mu,t}(x; R, T). \quad (2.34)$$

Thus the values written to the output file are already normalized by the color factor, averaged over the full lattice volume, and averaged over spatial directions.

In the implementation, the rectangular loop is constructed from two paths that connect opposite corners of the rectangle. The first path is

$$l_\mu = U_\mu(x; L_\mu) U_\nu(x + L_\mu \hat{\mu}; L_\nu), \quad (2.35)$$

and the second path is

$$l_\nu = U_\nu(x; L_\nu) U_\mu(x + L_\nu \hat{\nu}; L_\mu). \quad (2.36)$$

The measured loop contribution is then

$$\text{Re tr} \left(l_\mu l_\nu^\dagger \right). \quad (2.37)$$

This is equivalent to the trace of the ordered product around the rectangular closed contour.

The temporal Wilson-loop output file has the structure

```
# step, L, T, W_temp
```

where L is the spatial extent, T is the temporal extent, and W_temp is the averaged, normalized value defined in Eq. (2.33).

2.3.2. Effective Potential Extraction

Temporal Wilson loops are used to extract the static quark–antiquark potential. For large Euclidean time extent T , the Wilson loop has the spectral form

$$W(R, T) = \sum_n c_n(R) e^{-aE_n(R)T}, \quad (2.38)$$

where $E_0(R) = V(R)$ is the static ground-state potential and the higher $E_n(R)$ describe excited states with the same static sources. At sufficiently large T , the ground state dominates:

$$W(R, T) \propto e^{-aV(R)T}. \quad (2.39)$$

A lattice effective potential is therefore defined by the ratio

$$aV_{\text{eff}}(R, T) = -\ln \left[\frac{W(R, T+1)}{W(R, T)} \right]. \quad (2.40)$$

If excited-state contributions are negligible, this approaches the desired static potential in lattice units:

$$aV_{\text{eff}}(R, T) \xrightarrow{T \rightarrow \infty} aV(R). \quad (2.41)$$

In the numerical analysis, $W(R, T)$ denotes the already normalized and spatially averaged observable stored as `W_temp`. No additional color, volume, or spatial-direction normalization is applied during the analysis. The main remaining analysis choices are the thermalization cut, the treatment of autocorrelations, the temporal range used for the extraction, and the estimation of statistical uncertainties.

2.3.3. Creutz Ratios

Creutz ratios are combinations of neighboring Wilson loops designed to cancel some perimeter and normalization contributions. In the continuum they can be viewed as the mixed derivative

$$\chi(r, t) = -\frac{\partial^2}{\partial r \partial t} \ln W(r, t), \quad (2.42)$$

where $W(r, t)$ is a rectangular Wilson loop of physical spatial extent r and Euclidean time extent t . A central finite-difference discretization gives a lattice Creutz ratio located at half-integer separations,

$$\chi\left(R + \frac{1}{2}, T + \frac{1}{2}\right) = \ln \left[\frac{W(R+1, T)W(R, T+1)}{W(R, T)W(R+1, T+1)} \right], \quad (2.43)$$

or, equivalently after a shift of indices,

$$\chi(R, T) = -\ln \left[\frac{W(R, T)W(R-1, T-1)}{W(R, T-1)W(R-1, T)} \right]. \quad (2.44)$$

The normalization of the Wilson loop, including the factor $1/N_c$, cancels in this ratio. This is useful for the present $SU(2)$ analysis because the same formula can be applied directly to the normalized Wilson-loop output described above.

The central-difference form approximates the continuum mixed derivative with cutoff effects of order a^2 when the underlying Wilson loop is smooth. This makes Creutz ratios particularly useful for continuum studies: they are finite observables, their leading normalization and perimeter contributions cancel, and they are closely related to the static force. In the large-time limit,

$$\frac{\chi(R, T)}{a^2} \xrightarrow{T \rightarrow \infty} F(r) \quad (2.45)$$

up to the usual distinction between the discrete lattice midpoint and the physical distance r . Diagonal ratios $\chi(R, R)$ are also useful summary observables because they probe the same spatial and temporal scale and are straightforward to compare across ensembles [7, 6].

If large Wilson loops obey an area law,

$$W(R, T) \sim \exp[-\sigma a^2 RT - \text{perimeter terms}], \quad (2.46)$$

then the Creutz ratio approaches

$$\chi(R, T) \xrightarrow{R, T \rightarrow \infty} \sigma a^2. \quad (2.47)$$

Thus Creutz ratios provide a direct lattice estimator for the string tension in lattice units.

At finite R and T , Creutz ratios contain corrections from excited states, Coulombic contributions, finite-volume effects, and lattice artifacts. They are therefore best interpreted either as approximate string-tension estimators at large loop sizes or as diagnostic observables for scaling comparisons between ensembles.

For continuum extrapolations the important distinction is between fixed lattice separation and fixed physical distance. Holding R fixed while taking $a \rightarrow 0$ drives the physical distance $r = Ra$ to zero and therefore probes the short-distance limit rather than the same physical force. The continuum comparison should instead use a dimensionless distance such as

$$\hat{r} = \frac{r}{r_0} \quad (2.48)$$

and interpolate the measured Creutz ratios to common $\hat{r}$ values before fitting in a/r_0 . This is also where gauge-breaking effects are expected to be most visible: short-distance Wilson-loop observables are sensitive to ultraviolet deformations, while large loops suffer from rapidly increasing statistical noise.

2.3.4. Sommer Scale and String Tension

The static potential can also be used to set the scale. A common phenomenological form is the Cornell-type potential

$$V(r) = V_0 - \frac{\alpha}{r} + \sigma r, \quad (2.49)$$

where V_0 is an additive self-energy, α parametrizes the short-distance Coulombic part, and σ is the string tension. The corresponding force is

$$F(r) = \frac{dV}{dr}. \quad (2.50)$$

The additive constant V_0 drops out of the force.

The Sommer scale r_0 is defined implicitly by

$$r_0^2 F(r_0) = 1.65. \quad (2.51)$$

This definition uses the force at intermediate distances, where the potential can often be determined more accurately than at very large distances. In lattice units, the force is estimated from finite differences of the potential. If $\hat{V}(R) = aV(Ra)$, then

$$\hat{F}\left(R + \frac{1}{2}\right) = \hat{V}(R+1) - \hat{V}(R) \quad (2.52)$$

is a dimensionless estimate of $a^2 F(r)$ at an intermediate distance. The Sommer condition can then be written as

$$\left(\frac{r}{a}\right)^2 \hat{F}\left(\frac{r}{a}\right) = 1.65. \quad (2.53)$$

Solving this equation gives r_0/a .

The ratio r_0/a is one of the main scale-setting quantities in this thesis. It allows data at different β values to be compared in approximately physical units and provides the continuum variable

$$\frac{a}{r_0} = \left(\frac{r_0}{a}\right)^{-1}. \quad (2.54)$$

The continuum limit corresponds to

$$\frac{a}{r_0} \rightarrow 0. \tag{2.55}$$

The string tension may also be expressed in dimensionless form as

$$\sigma r_0^2. \tag{2.56}$$

This quantity is useful because it is independent of the arbitrary choice of lattice units. Agreement of such dimensionless quantities along a sequence of lattice spacings is a central test of scaling toward a common continuum limit.

3. Small Breaking of Gauge Invariance

3.1. Definition of the Gauge-Breaking Deformation

Define the modified lattice action or update rule used in the simulations. Specify the breaking parameters, for example ε_1 and ε_2 , and explain which gauge transformations are no longer exact symmetries.

3.2. Symmetry and Scaling Expectations

3.2.1. Relevant, Marginal, and Irrelevant Breaking Terms

3.2.2. Possible Restoration of Gauge Invariance

3.2.3. Expected Dependence on Breaking Strength and Lattice Spacing

3.3. Working Hypotheses

List the concrete hypotheses tested numerically:

1. Gauge-invariant observables approach the same continuum values for sufficiently small ε .
2. Gauge-breaking effects scale to zero with a measurable power of a or ε .
3. Above a threshold in ε , continuum extrapolations become unstable or incompatible with the gauge-invariant limit.

4. Numerical Setup

4.1. Simulation Parameters

Describe the gauge group, lattice volumes, coupling range, update algorithm, thermalization strategy, number of sweeps, and stored observables.

4.2. Ensemble Organization

Explain how ensembles are grouped by common physical and algorithmic parameters. State which parameters are varied in the continuum study and which are held fixed.

4.3. Thermalization and Autocorrelations

4.3.1. Thermalization Cuts

4.3.2. Integrated Autocorrelation Time

4.3.3. Bootstrap Blocking

4.4. Analysis Pipeline

Describe the analysis code and reproducibility workflow. Include the path from raw measurements to final observables:

1. Load simulation metadata from `input.yaml`.
2. Read Wilson-loop and observable measurements.
3. Apply thermalization cuts.
4. Estimate uncertainties with bootstrap resampling.
5. Extract $V(R)$, $\chi(R, T)$, r_0 , and related quantities.
6. Aggregate ensembles for continuum and breaking-strength comparisons.

5. Extraction of Physical Observables

5.1. Wilson Loop Analysis

5.1.1. Effective Mass Plateaus

5.1.2. Choice of Fit Windows

5.1.3. Systematic Uncertainty from Fit Ranges

5.2. Static Potential

5.2.1. Potential Extraction

5.2.2. Cornell-Type Fits

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5.3.3. Relation to the String Tension

5.4. Scale Setting

5.4.1. Determination of the Sommer Scale in Lattice Units

5.4.2. Alternative Scale Estimates

5.4.3. Consistency Checks

6. Continuum Extrapolation Method

6.1. Choice of Continuum Variable

The primary continuum variable is chosen to be the measured dimensionless lattice spacing

$$\hat{a} = \frac{a}{r_0}. \quad (6.1)$$

For observables that are expected to have leading even-power cutoff effects with the Wilson gauge action, the default horizontal axis is therefore $\hat{a}^2 = (a/r_0)^2$. Using the measured scale rather than the bare coupling makes the extrapolation closer to a fixed-physics comparison and reduces the risk of interpreting changes in scale setting as physical changes in the observable itself.

For Creutz-ratio observables the quantity to extrapolate should be dimensionless. Convenient choices are

$$\hat{\chi}(\hat{r}) = r_0^2 \frac{\chi(r, t)}{a^2}, \quad \hat{r} = \frac{r}{r_0}, \quad (6.2)$$

or, for large loops where the string-tension interpretation is appropriate, σr_0^2 . The measured lattice ratio $\chi(R, T)$ is dimensionless and approaches σa^2 in the area-law regime; division by a^2 and multiplication by r_0^2 converts it to a quantity with a finite physical continuum limit. The same logic applies to the force extracted from the static potential.

6.2. Fixed-Physics Trajectories

Ensembles at different β values are compared along trajectories where the physical volume and the dimensionless analysis point are kept as fixed as the available data allow. In practice this means comparing observables at common $\hat{r} = r/r_0$ and using only ensembles whose volumes are large enough that finite-volume corrections are not expected to dominate the Wilson-loop signal. When exact matching is not possible, the mismatch is included as a systematic effect or avoided by restricting the fit range.

The gauge-breaking parameters are treated as additional bare deformation parameters. If they are held fixed while a changes, the continuum fit tests whether their effect vanishes dynamically or survives in the continuum intercept. If a theoretical scaling prescription for ε is used, for example $\varepsilon \propto a^q$, the fit must instead include this scaling explicitly. The central diagnostic is whether the broken and unbroken data approach a common continuum value.

6.3. Fit Ansatz

6.3.1. Gauge-Invariant Baseline

For the gauge-invariant ensembles, a minimal continuum ansatz for a dimensionless observable $\hat{O}$ is

$$\hat{O}(\hat{a}) = O_0 + c_a \hat{a}^2 + c_{a4} \hat{a}^4 + \dots . \quad (6.3)$$

The $\hat{a}^4$ term is included only when the data support it or when it is used as a fit-systematic variation. The default fit should be stable under removing the coarsest lattice spacing and under modest changes of the physical-distance range used for the observable.

6.3.2. Combined Lattice-Spacing and Breaking Dependence

A useful shared-intercept test is

$$\hat{O}(\hat{a}, \varepsilon) = O_0 + c_a \hat{a}^2 + c_\varepsilon \varepsilon \hat{a}^p + c_{a\varepsilon} \varepsilon \hat{a}^{p+2} + \dots , \quad (6.4)$$

with $p > 0$. This ansatz encodes the hypothesis that gauge breaking is an irrelevant cutoff effect: for any fixed small ε the breaking terms vanish as $\hat{a} \rightarrow 0$, and all data share the same continuum intercept O_0 .

The competing separate-intercept test is

$$\hat{O}(\hat{a}, \varepsilon) = O_0 + c_\varepsilon \varepsilon + c_a \hat{a}^2 + c_{a\varepsilon} \varepsilon \hat{a}^2 + \dots . \quad (6.5)$$

Here a statistically significant c_ε indicates that the breaking deformation changes the continuum value of the observable rather than merely changing cutoff effects. If the deformation can enter symmetrically in the sign of ε , a quadratic term $c_{\varepsilon^2} \varepsilon^2$ should be included as a systematic variant.

This structure follows the same logic used in controlled continuum extrapolations with smearing or flow: one separates changes in the path toward the continuum from changes in the continuum observable itself [6]. In that setting, fixed smearing in lattice units can share the same continuum intercept if it remains in the controlled expansion regime, while a fixed physical smoothing radius defines a different continuum observable. In the present gauge-breaking problem there is no guarantee of a shared intercept, so Eqs. (6.4) and (6.5) are both part of the analysis.

6.4. Uncertainty Budget

6.4.1. Statistical Errors

Statistical errors are obtained from blocked bootstrap samples. The block size is chosen large enough to reduce residual autocorrelation effects in the saved Markov-chain history. Derived observables such as Creutz ratios, Sommer scales, and continuum-fit parameters are recomputed inside the bootstrap whenever possible, so nonlinear error propagation is captured by the resampling.

6.4.2. Fit-Window and Interpolation Systematics

Wilson-loop observables depend on analysis choices: the temporal range used for effective-potential plateaus, the loop sizes included in Creutz-ratio summaries, and the interpolation used to compare fixed physical distances. These choices are varied to estimate systematic effects. For Creutz ratios, short distances are most vulnerable to cutoff and breaking artifacts, while long distances are most vulnerable to statistical noise.

6.4.3. Finite-Volume and Continuum-Fit Systematics

Finite-volume effects are monitored by comparing ensembles with similar $\hat{a}$ and ε but different physical volumes when available. Continuum-fit systematics are estimated by changing the fit ansatz, removing the coarsest lattice spacing, and checking whether the extrapolation remains monotonic over the fitted range. Non-monotonic behavior or strong shifts of the intercept under reasonable fit variations are treated as evidence that the available lattice spacings are not yet in a controlled scaling regime.

7. Results

7.1. Thermalization and Data Quality

Insert representative thermalization histories and autocorrelation estimates.

7.2. Gauge-Invariant Baseline

Present the $\varepsilon = 0$ results for $W(R, T)$, $V(R)$, r_0 , and σ . This establishes the reference continuum behavior.

7.3. Dependence on Gauge-Breaking Strength

7.3.1. Observable Shifts at Fixed Coupling

7.3.2. Scaling of Shifts with Breaking Strength

7.3.3. Thresholds or Nonlinear Behavior

7.4. Continuum Limit at Fixed Breaking

7.4.1. Continuum Extrapolations for Each Breaking Strength

7.4.2. Comparison with the Gauge-Invariant Limit

7.4.3. Stability Under Fit Variations

7.5. Combined Scaling Analysis

Fit the dependence on both lattice spacing and gauge-breaking parameter. State whether the data support restoration of gauge invariance in the continuum limit.

7.6. Finite-Volume and Systematic Checks

Collect checks that are not part of the primary result but constrain the interpretation.

8. Discussion

8.1. Interpretation of the Continuum Limit

Discuss whether small gauge breaking behaves like an irrelevant lattice artifact, a relevant deformation, or a source of uncontrolled continuum bias.

8.2. Comparison with Theoretical Expectations

Relate the numerical findings to the hypotheses from Chapter 3.

8.3. Limitations

Discuss limits from statistics, accessible lattice spacings, finite volume, fit-systematic control, and the specific form of gauge breaking studied.

8.4. Implications

Explain what the result means for simulations or formulations where gauge invariance is only approximate.

9. Conclusion and Outlook

9.1. Summary of Findings

State the central quantitative conclusions.

9.2. Answer to the Research Question

Give the direct answer to whether the continuum limit is robust under the investigated small breaking of gauge invariance.

9.3. Future Work

Possible extensions:

- Study additional forms of gauge-breaking terms.
- Increase statistics and extend to finer lattice spacings.
- Add gauge-variant diagnostics to directly measure symmetry restoration.
- Compare different gauge groups or dimensions.

A. Simulation Tables

Provide complete tables of ensembles, including β , lattice size, breaking parameters, number of sweeps, thermalization cuts, and measured scales.

B. Additional Plots

Include supplementary Wilson-loop plateaus, bootstrap block-size scans, potential fits, and continuum-fit variants.

C. Analysis Code and Reproducibility

Document the exact command-line workflow, cache version, software environment, and data-selection rules used for the final analysis.

D. Derivations

Collect longer derivations that would interrupt the main text, such as Creutz-ratio identities or continuum-scaling arguments.

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